SIMATS ENGINEERING

CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

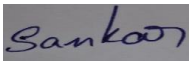
Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**

Question Count: 5

Code of Conduct

I kadiri sankar narayana (192325085) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Compare Type-1 and Type-2 Hypervisors for a Cloud Datacenter Hosting Mission-Critical Applications

Introduction

A hypervisor is software that creates and manages Virtual Machines (VMs). Hypervisors are classified into

1 Bare-metal

2 Hosted.

Hypervisor	**Hypervisor**
Runs directly on the hardware.	Runs on top of a host operating system.
Gives better speed and performance.	Performance is slower because it depends on the host OS.
More secure since there is no extra operating system layer.	Less secure because if the host OS has a problem, the VMs may also be affected.
Used in cloud data centers and enterprises.	Mostly used for learning, testing, and personal use.
Example: VMware ESXi, Hyper-V, KVM.	Example: VirtualBox, VMware Workstation.

Analysis

Performance

hypervisors access hardware directly, reducing latency and improving CPU, memory, and storage performance.

hypervisors depend on the host operating system, adding overhead and reducing efficiency.

Security

1 has a smaller attack surface because there is no host operating system.

2 inherits vulnerabilities from the host OS, making it more susceptible to attacks.

Manageability

Type-1 supports advanced features such as:

- Live migration
- High availability
- Fault tolerance
- Centralized monitoring

Type-2 is easier to install but lacks enterprise management capabilities.

Justification

- For mission-critical cloud applications, Type-1 hypervisors are the better choice because they provide:
- Better performance
- Stronger security
- Higher reliability
- Enterprise-grade management features
- Minimal downtime

2 A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

Introduction

Cloud providers often host multiple organizations on the same physical servers. Virtualization ensures that each tenant remains isolated.

How Virtualization Supports Isolation

Virtual Machines (VMs)

- Each department runs inside its own VM.
- Separate operating systems prevent interference.

Hypervisor Isolation

- The hypervisor allocates CPU, memory, and storage separately.
- One VM cannot directly access another VM's resources.

Virtual Networks

- VLANs or SDN create isolated network segments.
- Finance traffic remains separate from healthcare and student traffic.

Storage Virtualization

- Each tenant receives separate virtual disks.
- Access permissions prevent unauthorized access.

Two Major Attack Surfaces

1. Hypervisor Attacks

- Malware exploits vulnerabilities in the hypervisor.
- May lead to VM escape, allowing attackers to access other VMs.

2. VM Escape / Inter-VM Attacks

- An attacker gains access from one VM to another.
- Can expose confidential financial or healthcare data.

Conclusion

Strong hypervisor security, network segmentation, encryption, and regular patching are essential to maintain secure workload isolation.

3. VM Host Utilization Analysis

Given Snapshot

CPU Utilization = 92%

Memory Utilization = 88%

Storage I/O Latency = 35 ms

Average VM Boot Time = 170 seconds

Analysis of Bottlenecks

1. High CPU Utilization (92%)

- CPU resources are nearly exhausted.
- Causes slower VM execution and increased response time.
- Corrective Action
- Add additional CPU cores.
- Balance workloads across hosts.
- Migrate some VMs using live migration.

2. High Memory Utilization (88%)

- Memory pressure may cause swapping.
- Swapping significantly slows VM performance.
- Corrective Action
- Increase physical RAM.
- Reduce memory overcommitment.
- Allocate memory more efficiently.

3. Storage I/O Latency (35 ms)

- Normal latency is usually below 10 ms.
- High latency indicates storage bottlenecks.
- Corrective Action
- Upgrade to SSD/NVMe storage.
- Use storage caching.
- Separate heavy I/O workloads.

4. VM Boot Time (170 s)

- Long boot times indicate overloaded CPU and storage.
- Storage latency delays operating system startup.
- Corrective Action
- Improve storage performance.
- Optimize VM images.
- Increase available compute resources.

Conclusion

The primary bottlenecks are:

- CPU saturation
- Memory pressure
- Slow storage subsystem

These issues can be resolved by scaling hardware resources, balancing workloads, and improving storage performance.

4. How Software Defined Datacenter (SDDC) Improves Agility

Introduction

A Software Defined Datacenter (SDDC) virtualizes compute, storage, and networking, allowing all infrastructure to be managed through software.

Traditional Datacenter

- Hardware configured manually
- Slow provisioning
- Limited scalability
- High operational cost

Software Defined Datacenter

1. Compute Virtualization

- Physical servers host multiple VMs.
- New servers can be deployed within minutes.
- Resources scale dynamically.

2. Storage Virtualization

- Physical storage is pooled into logical storage.
- Capacity can be allocated on demand.
- Supports automated backup and disaster recovery.

3. Network Virtualization

- Software Defined Networking (SDN) creates virtual networks.
- Network configuration becomes automated.
- Policies can be changed without replacing hardware.

comparison

Traditional Datacenter	Software Defined Datacenter (SDDC)
Hardware-based management	Software-based management
Manual configuration	Automated configuration
Slow resource provisioning	Fast resource provisioning
Limited flexibility	Highly flexible and scalable
Lower resource utilization	Better resource utilization

Conclusion

- SDDC improves agility by enabling:
- Faster deployment
- Automatic scaling
- Better resource utilization
- Lower operational costs
- Simplified management

5. Identity Federation in a Multi-Cloud Environment

Problem Analysis

A multi-cloud federation project fails because each cloud provider uses different identity systems.

Problems include:

- Multiple user accounts
- Separate authentication methods
- Inconsistent access control
- Poor user experience
- Increased security risks

Secure Identity and Privacy Design

1. Identity Federation

- Establish trust between cloud providers.
- Users authenticate once and access multiple cloud services.

2. Single Sign-On (SSO)

- Users log in one time.
- Authentication is shared securely across providers.

3. Federated Identity Protocols

- Use industry standards such as:
- SAML
- OAuth 2.0
- OpenID Connect (OIDC)

4. Role-Based Access Control (RBAC)

Assign permissions according to user roles.

Example:

- Finance staff → Financial systems only
- Healthcare staff → Medical records only
- Students → Student portal only

5. Privacy Protection

- Encrypt identity information.
- Share only required user attributes.
- Apply Multi-Factor Authentication (MFA).
- Maintain audit logs for compliance.

Benefits

- Single Sign-On across cloud providers
- Strong authentication
- Improved privacy
- Consistent access control
- Better compliance and auditing

Conclusion

A secure federation architecture using SAML/OAuth/OpenID Connect, SSO, RBAC, MFA, encryption, and minimal attribute sharing ensures secure and seamless identity management across multiple cloud providers.